D4.1 Natural selection

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-d4-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 779–789 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

How selection works

D4.1.1 Natural selection drives evolution

pp. 779–781

Natural selection operates all the time and has done so for billions of years, producing the biodiversity of life on Earth. Individuals vary, more are born than can survive, and those better adapted leave more offspring.

D4.1.2 Mutation and sexual reproduction create variation

p. 782

Mutation makes new alleles, and **sexual reproduction** makes new combinations of alleles, through crossing over, random orientation and random fertilization.

D4.1.3 Overproduction and competition

pp. 782–783

More offspring are produced than resources can support, so members of a species compete. Food, water, space and nesting sites can limit a population's **carrying capacity**.

D4.1.4 Abiotic factors as selection pressures

p. 783

Density-independent factors, such as high or low temperatures, reduce survival at any population density. Individuals with heritable traits suited to the conditions are more likely to survive.

D4.1.5 Differences in adaptation, survival and reproduction

pp. 783–785

This is the basis of selection. **Fitness** is how likely a genotype is to survive and leave offspring, and it depends on the environment. Competition within a species favours those with higher fitness.

D4.1.6 Traits must be heritable

p. 784

Characteristics acquired in life because of the environment are not encoded in base sequences, so they are not inherited and cannot evolve.

Sexual selection

D4.1.7 Sexual selection

pp. 786–788

Physical and behavioural differences can affect success in attracting a mate, and they can signal fitness. Birds-of-paradise are well-known examples, in which males have exaggerated plumage and displays.

D4.1.8 Modelling selection: Endler's guppies

pp. 788–789

Interpreting data from a guppy experiment

D4.1.8

Pond	At start	5 months	14 months
No predators	10.0	11.5	12.5
Weak predators	10.0	11.0	12.0
Dangerous predators	10.0	9.0	8.5

Mean spots per male. Simplified from the pattern in Endler's work, for practice.

Activity

Describe each trend and explain it with natural and sexual selection. Predict what happens if males from the dangerous pond move to a predator-free pond.

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Bright spots attract females, as well as predators. Endler kept guppies under different levels of predation. Spots increased without predators, through sexual selection, and fell with dangerous predators, through natural selection. Selection pressures in the experiment were controlled.

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